Manufacturing Execution System for Titagarh Rail Systems

Powered by



Date: July 2, 2025

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EXECUTIVE SECTION

Proposal Cover Letter

To,
Mr. Mayukh Saha,
Titagarh Rail Systems
Kolkata, India

Dear Mr. Mayukh,

We are pleased to submit the following proposal to Titagarh Rail Systems (TRSL) for Manufacturing Execution System for your existing plant operations. We hope and believe, this document provides you with all the information you sought for. Should you require additional information or need clarification, please do not hesitate to contact us.

As a dedicated market leader in Manufacturing Operations Management solutions, we are confident in our ability to deliver an integrated production and operations management and tracking system that will provide TRSL with a flexible and scalable platform for growth and continuous improvements.

Thank you for giving us an opportunity to be a potential supplier to TRSL for your upcoming projects.

Rahul Soni
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Dassault Systèmes India Pvt. Ltd.
Pramuk MM Meridian, 5th & 6th floor
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Bengaluru, Karnataka, 560070, India

Executive Summary

Dassault Systèmes is pleased to submit a formal response for implementation of Manufacturing Execution System for TRSL's rolling stock assembly plant at Kolkata. We are confident in our ability to deliver a completely integrated solution that will address your needs and provide a comprehensive platform to satisfy many long-term business objectives. **3D**EXPERIENCE platform for Manufacturing is proven in helping TRSLs effectively meet increasing volume and complexity demands while simultaneously improving quality, compliance, and Return on Assets (ROA).

Our confidence in our ability to deliver is based on the solution capabilities and consulting and implementation experience. Our solution for supply chain & operations management for manufacturing has successfully deployed solutions to 180+ customers across 40+ countries and more than 1400+ factories worldwide. Our platform is the industry-recognized leader for industrial operations management solutions for complex discrete, batch process, and hybrid production process customers. We have proven ability to get systems operational on time and on budget. For over 20 years, we have delivered enterprise-wide supply chain & operations management systems that provide **visibility into, control over, and synchronization across** manufacturing operations. Our solution improves responsiveness and agility while addressing regulatory compliance requirements at the lowest total cost of ownership.

Dassault Systèmes supports global implementations for a diverse range of customers in many industries including Automotive, Industrial Equipment, Aerospace and Defense, Consumer Goods, Energy & Materials and Life Sciences. Recognized for its comprehensive functionality, operational flexibility and global manufacturing capabilities, **3D**EXPERIENCE platform for supply chain & operations management is rapidly assuming a leadership position within the growing 'Clean-tech' sustainable energy products market. Tesla Motors, SCANIA, Boston Power, Showa Shell, Solar Frontier, Bloom Energy and GM's Chevy Volt battery assembly operation have all selected our platform to deliver their cutting-edge products to market efficiently and at the highest level of quality.

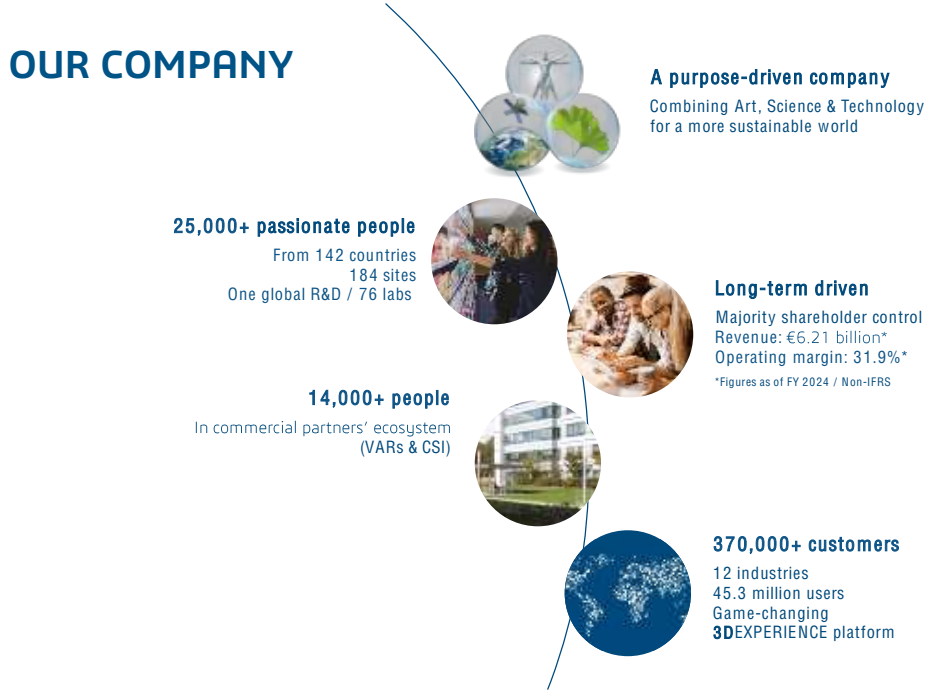
Our solution approach is built on the key themes of –

- Service transformation through automation and process improvement leveraging our framework and tools, which helps deliver cost reductions to TRSL.
- A balanced, risk-mitigated transition to a new platform that is possible because of our industrialized transition approach and understanding of the application portfolio
- Our commitment to help TRSL operations evolve not just with the technologies, but also with ease of operations over time, which will result in enhanced and optimized business processes.
- Rapid roll out of solution across multiple lines / facilities at TRSL through accelerated and standardized approach

Dassault Systèmes – Company Overview

Dassault Systèmes, the 3DEXPERIENCE Company, founded in 1981, and for more than 40 years, we have been providing businesses and people with virtual universes to imagine sustainable innovations. Its 3DEXPERIENCE Platform leverages the Company's world-leading 3D software applications to transform the way products are designed, produced, and supported, enabling businesses to craft delightful TRSL experiences.

With the 3DEXPERIENCE Platform, Dassault Systèmes customers create "social enterprises" that involve their customers in the innovation process. With its online architecture, the 3DEXPERIENCE environment helps businesses to test and evaluate — anywhere in the development lifecycle of a product or service — the eventual experience they will deliver to their customers. In short, 3DEXPERIENCE powers the next-generation capabilities that drive today's Experience Economy. 3DEXPERIENCE platform delivers best-in-class software applications that provide 3D Design, Engineering, 3D CAD, Modeling, Simulation, Data Management and Process Management, Digital Manufacturing, Manufacturing Operations Management, Sales & Operations planning



The 3DEXPERIENCE® platform

The 3DEXPERIENCE platform is a platform for knowledge and knowhow, a game-changer collaborative environment that empowers businesses and people to innovate in an entirely new way.



The 3DEXPERIENCE platform is symbolized by the compass and empowers customers & users to:

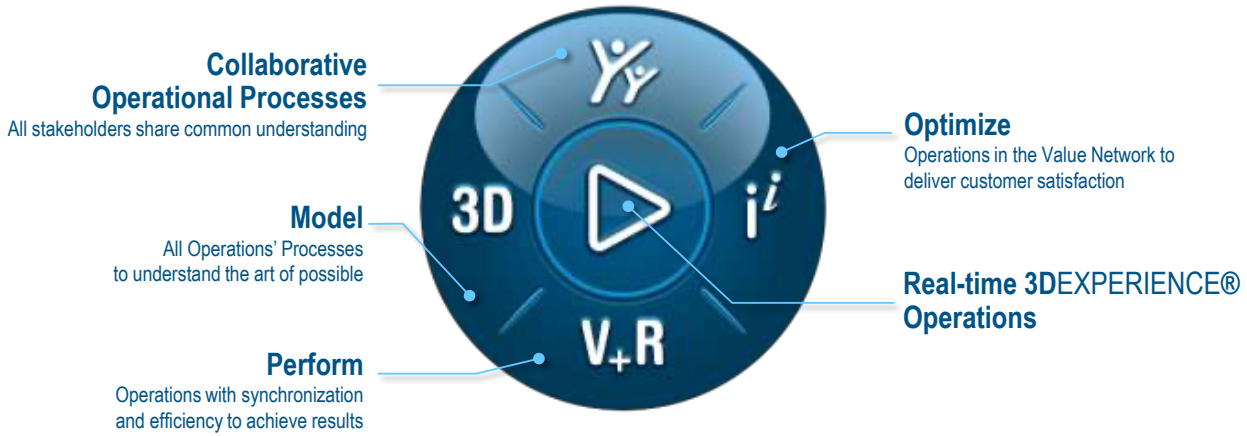
- create and visualize products and the related experiences (West quadrant)
- evaluate the possible business solutions by confronting them with reality (South quadrant)
- calibrate and contextualize business solutions considering all the information within and outside the company (East quadrant)
- bring together and catalyze a diversity of talents (North quadrant)

We have 370,000 enterprise customers and 45.3 million users across 12 industries, in 140 countries. Our customers range from major accounts to startups and governments, as well as individual users. We provide solutions for all sectors of the economy, from mass retail to high-tech.


Transportation & Mobility
Aerospace & Defense
Marine & Offshore
Industrial Equipment
High-Tech
Home & Lifestyle
Consumer Packaged Goods - Retail
Life Sciences
Energy & Materials
Construction, Cities & Territories
Business Services

3DEXPERIENCE for Manufacturing Operations Management – Delivered by Dassault Systèmes

An integral part of the Dassault Systèmes' 3DEXPERIENCE platform, is to provide the connection between the virtual and the real world. 3DEXPERIENCE for Operations Management helps global businesses re-imagine their planning, management and optimization of industrial operations.



In today's industry trend towards agile production processes, one of the key enablers is operational excellence. Operational excellence requires harmony across design, supply chain, production, distribution, people and processes. Our portfolio will enable TRSL to transform operations easily and safely by designing and testing them in a simulated production environment. Once completed, you can efficiently plan, produce, and manage all resources from staff, to production, to TRSL delivery to enable end-to-end visibility into manufacturing operations.

In addition, our platform allows all stakeholders in operations, no matter their level of expertise, to be part of a single community that has all of its members, and all of its assets, working toward the same-shared objectives of operational excellence.

DELMIA for Industrial Operations Management of the Future

The **DELMIA** platform for Global Industrial Customers help them to transform their global production operations to achieve and sustain operational excellence. **3D**EXPERIENCE platform provides visibility into, control over and synchronization across operations and processes on a global and multi-site scale.



By managing operations from an enterprise perspective, **3D**EXPERIENCE platform helps customers to improve operational flexibility, resulting in a greater ability to respond quickly to change. As a result, continuous process improvement initiatives, such as Lean or Six-Sigma programs, can yield greater results with sustained operational improvement. Further, the scope of continuous improvement can be effectively replicated across all operations on a global scale.

Furthermore, 3DEXPERIENCE platform solutions help global manufacturers stay ahead of the competition by delivering better visibility into, control over and synchronization across the business processes performed within and across sites and supplier partners. Customers can leverage **3D**EXPERIENCE platform to establish a common set of operational standards that can be managed holistically on a global scale while still being continuously improved to meet local market and customer needs resulting in better integration and real-time visibility across the global enterprise

operations. Customers aiming to grow revenue and expand into emerging markets must achieve these objectives while improving efficiency, removing waste, reducing cost and continuously improving quality. Balancing these goals, cost effectively is at the heart of manufacturing excellence.

Manufacturing Operations Management Platform – DELMIA Apriso

DELMIA Apriso solution is a global manufacturing platform that provides visibility into, control over and synchronization across manufacturing operations. Built as a unified application it delivers integrated processes leveraging a single data repository across production, quality, maintenance, logistics / warehouse, and time and labor.

It is a web-based system with a Services Oriented Architecture that can easily adapt to any manufacturing environment: simple or complex, small or large, high or low volume. It also seamlessly integrates with ERP, PLM and other business systems, enabling manufacturers to achieve the enterprise-wide agility required to respond to new market opportunities, changes in demand and shifts in strategy. Many customers deploy DELMIA Apriso as their Operations Portal for manufacturing personnel.



Figure 1. The Suite of DELMIA Manufacturing Applications

- ✓ DELMIA is a Global Solution that provides Visibility, Control and Synchronization
- ✓ Across All Operations (Production, Quality, Warehousing, Maintenance, Time & Labor),
- ✓ Based on a Leading Technology (Unified Architecture, Unified Processes, Unified User Interface, Unified data Model)

Digital Operations with DELMIA Apriso Paperless Manufacturing

DELMIA Apriso Paperless Manufacturing solution enables customers to achieve digital operations and paperless manufacturing. DELMIA Apriso Paperless Manufacturing is a proven, highly robust, and comprehensive solution to removing paper from operations. The solution brings the speed and agility of paperless across TRSL′ manufacturing operations while managing the complexities and details of the plant floor that are beyond the capabilities of ERP. From tight integration with automation systems, directed tasks synchronization, work instructions, to rapid handling of Engineering Change Orders (ECO) and product introductions, to manufacturing intelligence that drives continuous improvement, leading manufacturers are using DELMIA for Paperless Manufacturing to achieve new levels of efficiency, visibility, compliance and traceability throughout their enterprise.



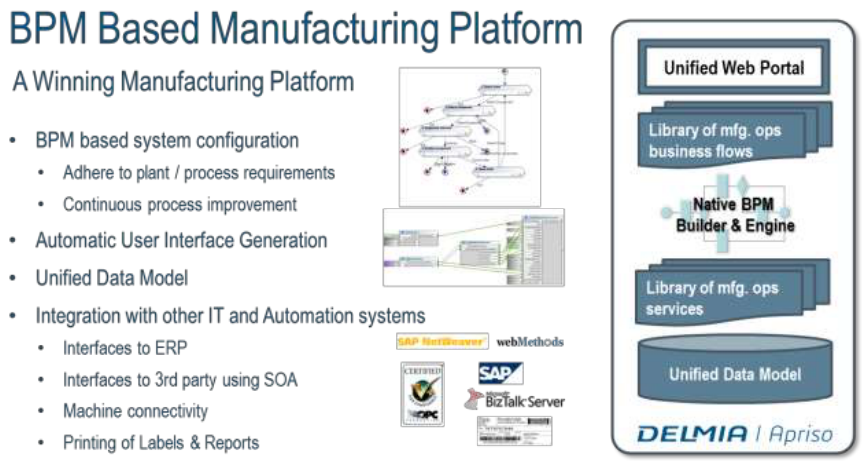
Figure 2 DELMIA Apriso for Paperless Manufacturing

DELMIA Apriso Paperless Manufacturing solution has the following benefits for TRSL:

- Accelerate NPI & Innovation
- Enable Material Synchronization
- Increase Manufacturing Efficiency
- Improve Quality & Compliance

Flexible Production with DELMIA Apriso Business Process Management

Global manufacturing leaders are those able to respond quickly to change while maintaining process consistency and quality standards. One approach they take is to establish a COE team to configure and distribute process improvements. By combining a COE with a global platform for manufacturing operations management that embeds Business Process Management (BPM), TRSL can simplify process distribution, governance and monitoring, resulting in late product and process customization and changes with less effort.



DELMIA offers a unique application that is based on cross-functional processes, which are deployed across manufacturing locations. Processes authored in DELMIA Process Builder reference all elements of manufacturing operations, including work instructions (WI), documents and characteristics, and specifications. These processes also include technical components, such as database objects, reports, label definitions, machine configuration and dictionary data. Hence, TRSL can easily and quickly push late changes in its processes (including WI, documents, etc.) to all its plants.

DELMIA Apriso BPM has the following unique capabilities:

- Single tool for authoring unique business processes.
- Supports Technical & non-technical users
- Supports integration to automation layer
- Supports integration to Business Systems (ERP, PLM, SCM, etc.)
- Supports run-time execution of business logic

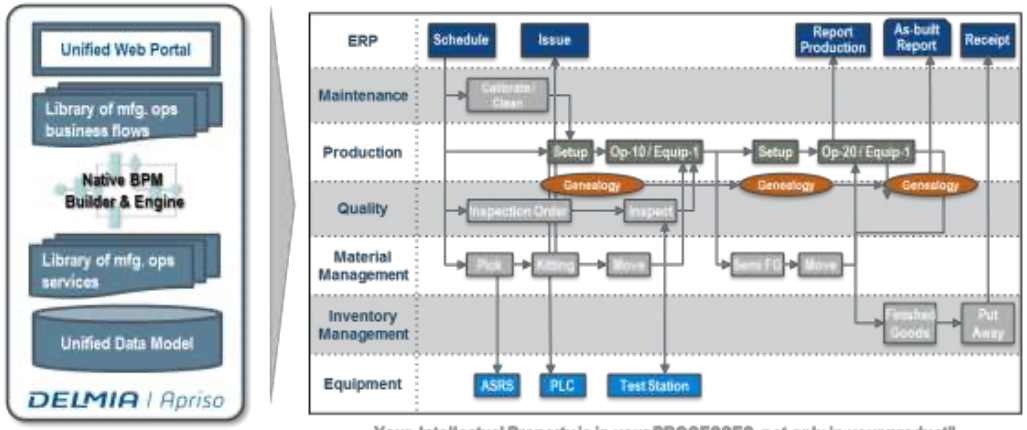
- Supports automatic generation and rendering of shop floor UI without coding
- Supports a broad scope of mobile devices
- Support of Prototyping - define only the "surface" layer of a process including UI, screens, buttons, and navigation between screens
- Supports Blueprinting - allows for in-line documentation of business processes

The native BPM technology is a key differentiator for DELMIA, which makes it more adaptive to the speed of change in two ways:

- By empowering plant-floor operations and information technology professionals to configure or reconfigure the system with a minimal amount of "configuration" and with no programming needed (no compilation, external sources, packaging).
- By coordinating end-to-end processes that synchronize all areas of operations, from materials, to production, quality, maintenance, and shipping.

Cross-Functional Unified Processes with DELMIA Apriso Unified Data Model

The DELMIA Apriso Unified Data Model (UDM) was built from the ground up using a single, integrated, and unified data modelling. In DELMIA Apriso whether it be a Resource (Machine, Equipment, Tools, and Labor), Lot, Serial, Material Characteristic, etc., there is only one definition for each. The UDM supports all execution and reporting across production, material, quality, maintenance and labor, in one plant or across the enterprise and they are cross-functionally linked, e.g., material ↔ order, quality / test results ↔ genealogy, and equipment used ↔ test results.



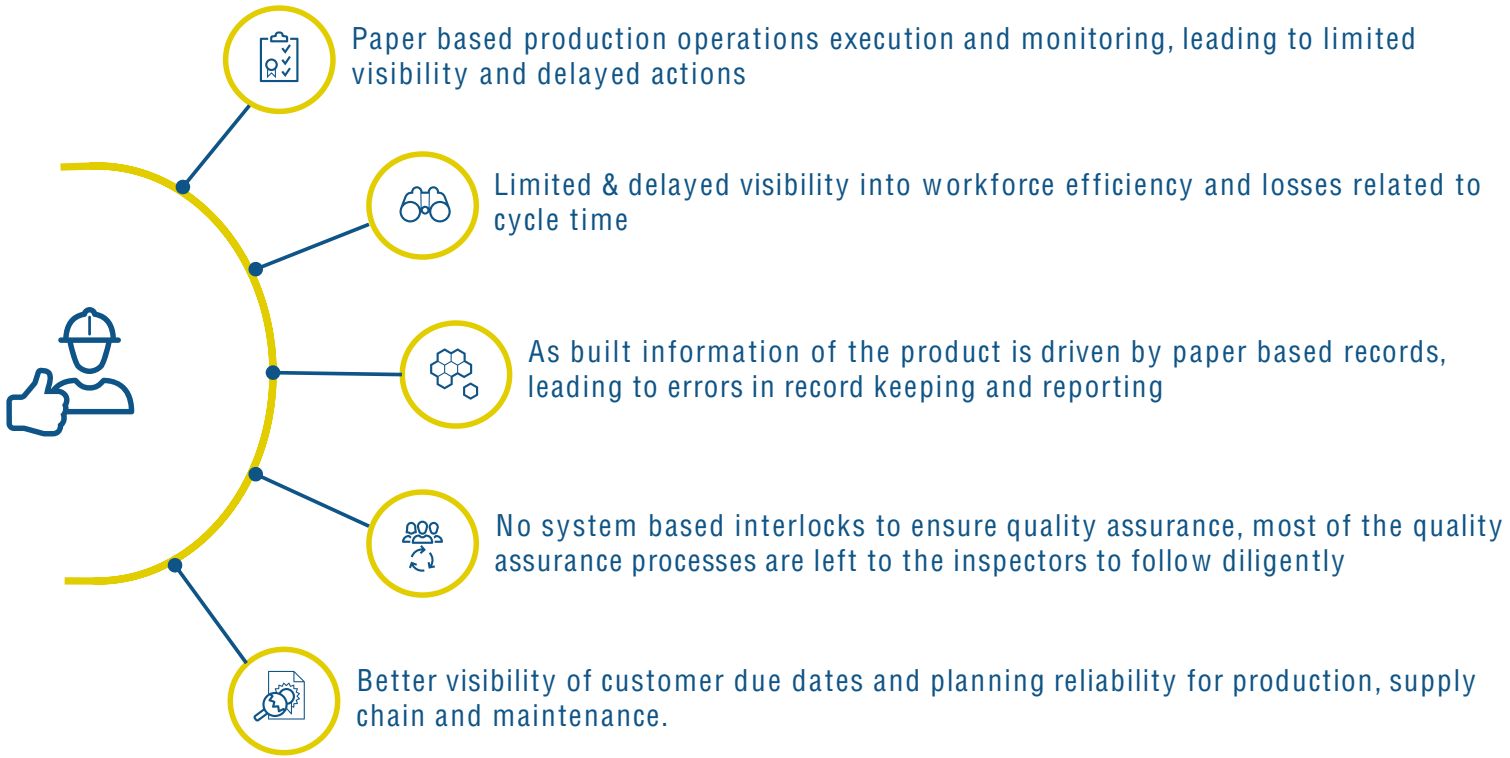
Your Intellectual Property is in your PROCESSES, not only in your product"

Using DELMIA Apriso UDM and process authoring capabilities, TRSL will be able to create cross-functional unified processes. These processes not only orchestrate the tasks related to different functional domains, but they can also operate on multiple domains in the same task.

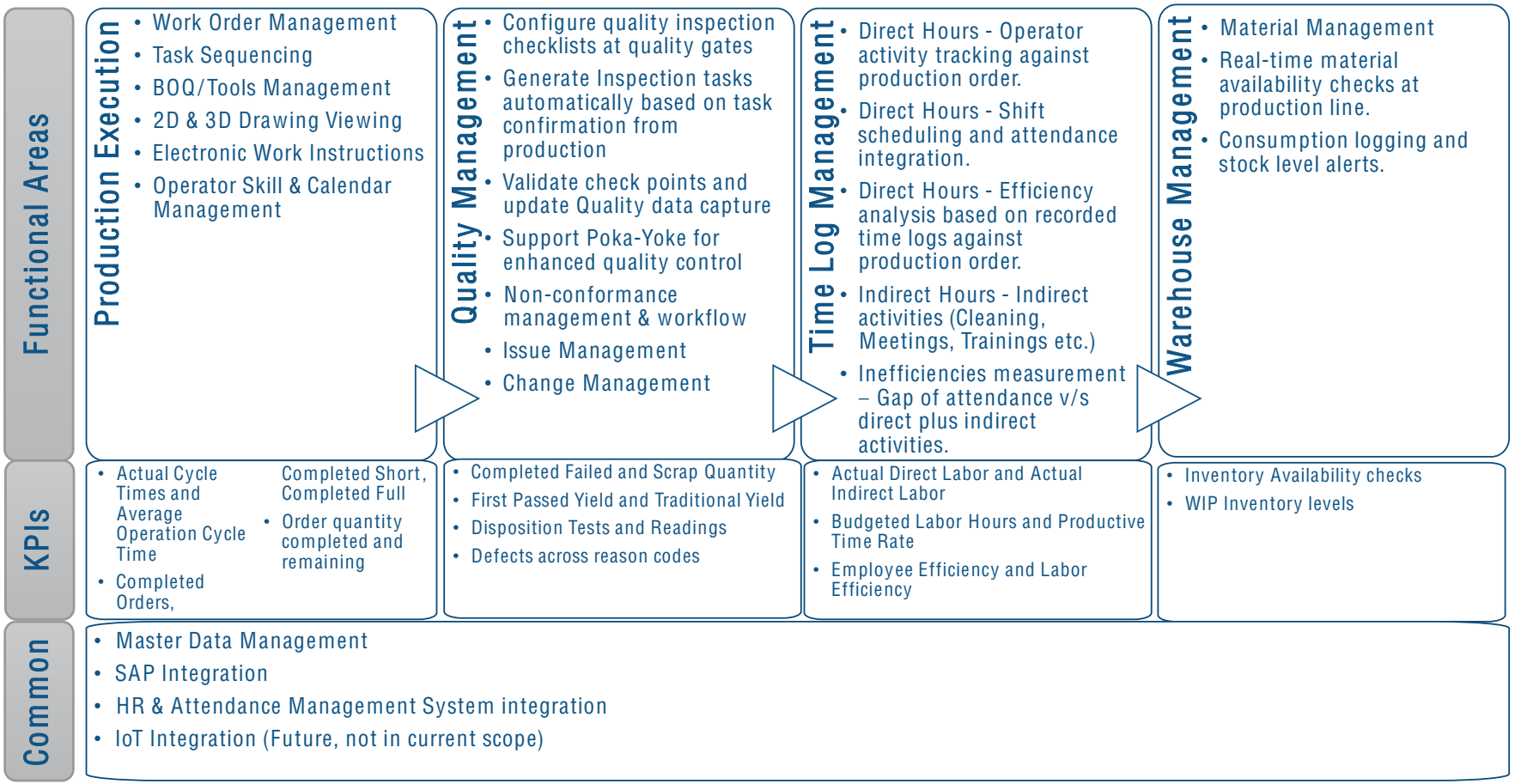
Our Understanding of TRSL'S Key Business Objectives & Requirements

TRSL is looking to leverage digital technologies for managing rolling stock manufacturing operations that will allow them to get better visibility into the production & quality activities along with the labor productivity details. This will allow TRSL to get better baseline of the labor / effort required to build / assemble rolling stock, which will be helpful during bidding process as well as identify areas of improvement and training for continuous improvement.

Below are the key challenges being faced by the operations team -



To address the production challenges, TRSL would like to evaluate the technical capabilities of DELMIA portfolio of Dassault Systemes to address the following areas:



Solution to TRSL MES Requirements

Based on our understanding of TRSL requirements, we are convinced that we are the right partner to help TRSL in implementing a MES solution to achieve the required level of production tracking & monitoring to meet the production targets and operational efficiency set out in the objectives of this program.

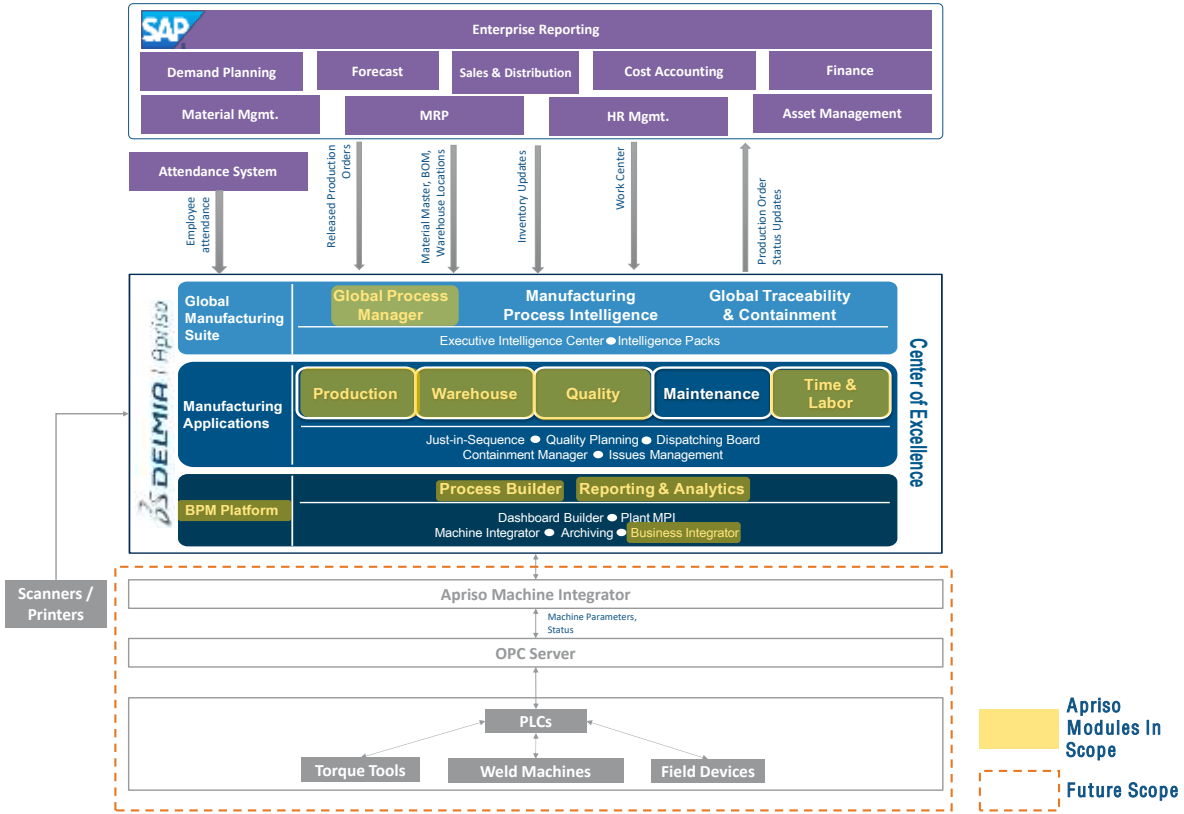
This TRSL initiative is to pave the way for a scalable MES solution for rolling stock production – improved operational efficiency, improved capacity, and improved levels of quality & safety. The DELMIA platform solutions components are fully aligned with TRSL's requirements and can help achieve this by providing necessary capabilities. In the following sections, we have summarized some of the platform's key features in alignment with TRSL's MES requirements.

In the current scope of this demonstration document, we are addressing TRSL's MES requirements by implementing a Vertically Integrated and Networked Operations Management System. The details of this solution are provided in the subsequent sections. To summarize, the solution will address TRSL's key objectives:

- Synchronized operations involving material and information flow across multiple manufacturing work centers
- Agility in operations, as the production process in rolling stock industry ranges is very detailed and involves multiple operations and long cycle time, the MES solution must be able to track production activities to better react to manufacturing events and make quick decisions.
- Flexible and scalable solution to meet continuous improvement and future initiative requirements.
- Ease of implementation and maintenance of the manufacturing IT landscape over the long run.
- Bring all the stakeholders and operational disciplines across all facilities together to have full visibility, control, and synchronization of manufacturing operations.
- Provide one single integrated platform to keep it simple and ensure everyone gets a unified view.
- Highlight bottlenecks and inefficiencies in processes; enable interventions to improve operational excellence.

Solution Coverage for TRSL

The DELMIA platform for Operations Management will integrate with existing SAP ERP systems and attendance systems in order to deliver an agile end-to-end IT landscape capability. Below is a high-level view of the integrated architecture involving SAP ERP and other IT & automation systems. The platform comes with interfaces with SAP ERP (and with multiple other ERPs). In addition, the platform come with capability to build additional configurable user interfaces as required, based on client specific business processes. It also consists of inherent capability to integrate with automation systems used in the field through the Machine Integrator module. Having said that, integration with shop floor automation systems is currently not in scope for implementation.



It is important to note that DELMIA platform is a modular application and any new module can easily be extended to TRSL in the future as the platform is built on unified data model and architecture. The table below summarizes the DELMIA Apriso modules compliance to RFP feature requirements –

DELMIA Modules / Functional Requirements	APS	Production	Warehouse	Quality	Quality Planning	Issue / Containment Management	Maintenance	Time & Labor	Business Integrator	Machine Integrator	MPI Lite / Dashboard Builder	Process Builder / Execution Platform	Process Manager	Archiving	Prepackaged Solutions
Work Order Management		●							●						●
Task Sequencing		●							●						●
Time Log Management								●	●						
Issue Management				●					●						
Operator Skill & Calendar Management		●						●	●						●
BOQ/Tools Management		●													
2D & 3D Drawing Viewing		●							●						●
Reporting & Dashboards		●	●	●				●	●		●	●	●		
Master Data Management		●	●	●				●	●						
SAP Integration									●						
Quality Management & NCR				●											
Electronic Work Instructions		●													
Material Management			●												
Integration Capabilities									●						
Change Management		●	●												

Project Implementation Methodology

TRSL MES Scope Considerations – Functionalities and Divisions In Scope

IN SCOPE | PHASE 1 | METRO DIVISION

- Implementation of DELMIA Apriso with Time and Labor, Production, Complex assembly modules on TRSL Kolkata factory for Metro Division
- Configuration of master data in DELMIA Apriso for Metro Division
- DELMIA APRISO Modules
  - Time and Labor
  - Production
  - Quality
  - Warehouse
  - Complex Assembly
- Development of 10 Reports and 10 Dashboards.
- ERP Integration with Apriso
- Configure Metro Division in Apriso as per below table
- Go-Live and critical care

Shops	Line / Work Station	Work stations	Routing Steps or Operations	No of Projects	No. of Quality Gates	No. of Shifts
Metro	Bay 3/ Carbody	34	114	3	12	3
	Coldchamber	3	30	3	3	3
	Bay 2/ Outfitting	19	550	3	25	3
	Shower Test & Car Weighment	2	6	3	3	3
	Testing	3	20	3	20	3
	Bogie Shop	10	20	3	4	3
	Pipe Bending Facility	10	20	3	5	3

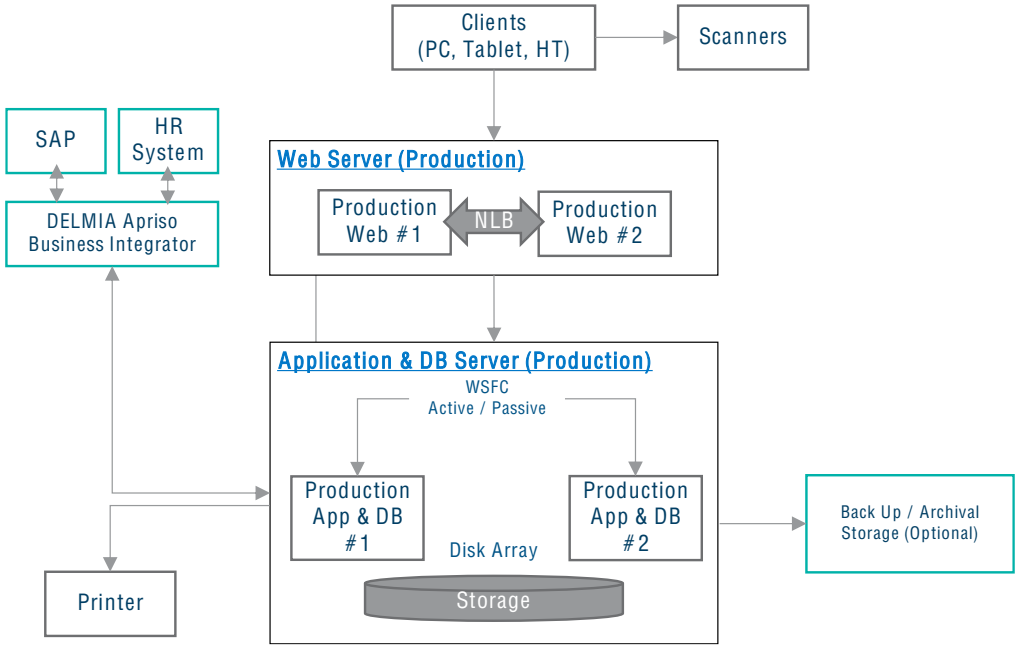
Our Solution Offering

Proposed Software Modules of DELMIA Apriso for Phase 1 (Metro Division)

Module Name	Description
Business Integrator	**Business Integrator** provides seamless integration points, which allow DELMIA Apriso to easily adapt to your existing IT infrastructure by using XML technologies. (SAP, Dassault Systèmes, etc.)
Execution Platform	**Execution Platform** is an operations execution platform, core component of the DELMIA Apriso Suite, and is required to run all functional software products.
Process Builder	**Process Builder,** which utilizes a native Business Process Management (BPM) framework, allows you to author and graphically define manufacturing processes spanning multiple operational areas such as production, warehouse, quality, maintenance and labor operations.
Production	**Production** provides visibility and directed real-time control of manufacturing operations, tracking work-in process, execution tasks, labor and machine resources, while supporting your continuous improvement, Lean manufacturing and Six Sigma initiatives.
Production Supervisor Role	The Production Supervisor Role is necessary to execute the following actions. A Production Supervisor is: A production employee responsible for monitoring and managing operations in the Production Monitoring Cockpit screen. An administrator responsible for setting all the configuration options in the Production Monitoring Cockpit Configuration screen.

Quality	**Quality** monitors and manages all aspects of production quality control and quality assurance execution. The Quality software products tracks SPC trends, finds defects and pinpoints their root causes at specific operational points. Quality can also trigger alerts for supervisors to take corrective action for process anomalies and dynamically diverts work to an alternate workflow. Statistical Process Control (SPC) performs the complex statistical analyses required by manufacturers and design engineers to streamline operations, improve overall quality and execute Lean and other continuous improvement initiatives. Visual Quality Defect Tracking tracks and analyzes product defects at the level of individual occurrence, including product configuration, runtime and analysis
Warehouse	**Warehouse** offers real-time, end-to-end visibility and control for optimizing your logistics operations. Tying together inbound and outbound logistics with warehouse operations, the Warehouse software products manages cross-docking, receiving, warehousing and shipping operations.
Time & Labor	**Time & Labor** is a browser-based, enterprise-wide portal for the real-time collection and management of time-based data.
Plant MPI aka MPI Lite & Dashboard Builder	**Plant Manufacturing Process Intelligence (MPI Lite)** is an out-of-the-box solution for the local plant, requiring no costly infrastructure or integration. Deployment is fast and easy, and users will be quickly productive as they work with the product's intuitive graphical interface to increase efficiency, improve quality, and reduce plant-wide manufacturing cost. It also comes with Dashboard Builder to help business users create dashboards by themselves
Process Manager	**Process Manager** lets you package, deploy and govern manufacturing processes across multiple plants, warehouses and other facilities. GPM deploys systems configuration data and processes with logic and user interface, database objects, machines configuration and third-party components.

Proposed Solution Architecture



Note – The proposed architecture is based on current requirement understanding for Metro division only and is subject to change post Blueprinting phase. The sizing for other divisions will be done post implementation of Metro division.

Proposed Hardware Sizing

Deployment Scenario	Server Class	Hardware Resources	Specs	OS Installed	Software Installed
Web Server with Network Load Balancing	2 x Web Node	CPU	6 vCPU	Microsoft Windows Server 2022 – only 64 bit	Internet Information Services (IIS) 64-bit version of IIS shipped with your OS, with the following server roles installed: ASP.NET 4.8. WCF Http Activation. Windows Authentication (only if Windows Integrated Authentication is to be used in DELMIA Apriso)
		Memory	16 GB		
		Local Storage	200 GB		
		NIC	2x 1Gbps NIC		
Application Failover Cluster Active / Passive	2 x App & Database Node	CPU	12 vCPU	Microsoft Windows Server 2022 – only 64 bit	Apriso Application Software and Services Microsoft SQL Server 2022
		Memory	128 GB		
		Local Storage	200 GB		
		NIC	2x 1Gbps NIC		
		DAS/SAN Storage	500 GB (1 year approximately)		
Backup / Archival Storage (Optional)	1 Server	H/W Sizing	6 Core / 16 GB RAM/ 200 GB Local Storage	Microsoft Windows Server 2022 – only 64 bit	Microsoft SQL Server 2022
		Storage	1TB		

Note – The above & below mentioned server sizing is based on preliminary understanding of process and data requirement for Metro shop. This will be further validated post Blueprinting phase and is subject to change. The sizing for the rest of the two divisions will be done post implementation of Metro division.

Scenario Name	Server role	No. of server	HW type	Apriso HW spec
DEV	Single server hosting all tiers	1	Virtual machine / Physical	6 vCPU 24 GB RAM 100 GB storage (local) 200 GB DB Store

				1x 1 Gbps NIC
TEST / QA - Apriso	Two Tier Architecture	2	Virtual machine / Physical	8 vCPU 32 GB RAM 100 GB storage (local) 300 GB DB Store 1x 1 Gbps NIC

Pre-requisites / assumptions

- 3rd party software such as Windows Operating System and SQL Server Database to be procured, installed & configured by TRSL
- Procurement, Installation and Configuration of 3rd party software such as OS, SQL Server, OPC Server, etc. will be in TRSL's scope
- Hardware for running all the requirement DELMIA Apriso components and various 3rd party software will be procured, installed and configured by TRSL
- Any updates to the shop floor automation systems to send data to MES layer will be enabled by TRSL
- The server sizing and architecture details are subject to change based on detailed discussions post Blueprinting phase

Proposed Commercials:

Budgetary Software Commercials: Phase wise Software Pricing:

- **Phase 1 – For Metro**

  For Metro considering 200 resources (200 resources in 2 shifts) for Production + Time and Labor + Quality + Warehouse modules pricing of INR **2,70,00,000/-** (PLC + 1-year ALC) + GST at actuals

Terms & Conditions:

- For 200 employees /resources considered across 2 shifts.
- The pricing is for one site (Titagarh, Uttarpara site). The modules included are defined in the earlier sections.
- This assessment is for Metro, in the current phase.
- The above pricing is inclusive of 1st year AMC
- Above commercials are exclusive of taxes
- Hardware cost is not included in the proposal.
- Final proposal will be based on the SoW signoff.

Terms and Conditions

- Prices are quoted in INR (₹).
- Purchase order to be placed as per the below-mentioned details.

  Dassault Systèmes India Pvt. Ltd.

  Pramuk MM Meridian, 5th & 6th floor

  No. 46/1, K. R. Road, Jayanagar 7th Block,

  Bengaluru, Karnataka, 560070, India
- Payment Terms: 30 days from the delivery date of the invoice.
- HSN: 997331.
- GST Registration Number - 29AACCD7672A8ZI
- Taxes – Extra as applicable at the time of delivery.
- The license agreement is as per DS CLOSA.
- Validity: July 31, 2025
- The license keys will be delivered in 15 working days after PO Release and the required documentation signing

Annexure – Response to the Additional Questions:

1. Software Platform

Below is the description of all the DELMIA Apriso modules that are being proposed for the solution at TRSL.

DELMIA Apriso Production

DELMIA Apriso is the only MES application of its kind with the broad functionality and flexibility to support today's extended manufacturing execution environment. This application provides visibility and directed real-time control of manufacturing operations, tracking work-in process, execution tasks, labor and machine resources. It enables paperless manufacturing by presenting both text and graphical electronic work instructions, validating input materials and components while collecting production data from both automation and human interface sources. It includes highly configurable electronic manufacturing records, which capture complete genealogy and interlocking product and process traceability across machines, work centers, tooling and people. DELMIA Production enables manufacturers to rapidly identify the root causes of operational issues and improve manufacturing processes across your production environment.

Specific capabilities include, but are not limited to:

- Capturing raw material and process traceability data for each unit or lot, including component serial or lot numbers); machines, operators, inspections, set points, rework and test results
- Supplier traceability, incorporating "as-built" data from feeder plants and other suppliers
- Customer traceability ensured through shipping history

DELMIA Production is an optimized application to support Lean and other Continuous Improvement initiatives to drive waste out and efficiency in to your production processes.

DELMIA Apriso Quality

It delivers an advanced quality management and execution capabilities by monitoring and managing all aspects of quality control and quality assurance execution. It enforces and executes the quality plan created in PLM, ERP or DELMIA Quality Planning systems and performs quality functions across a wide range of business processes going beyond production to include warehouse, supply chain, maintenance and labor business processes. Designed to be as automated, flexible and paperless as possible, DELMIA Quality tracks Statistical Process Control (SPC) trends, finds defects and pinpoints root causes at specific operational points. DELMIA Quality alerts supervisors to take corrective action for process anomalies,

dynamically diverting work to an alternate workflow if so desired. With DELMIA Quality non-conformance, Corrective Action Preventative Action (CAPA) can be linked into production for a highly efficient process that will continually improve your overall product and process quality.

By combining the directed, quality execution capabilities of DELMIA Quality with existing capabilities of ERP or PLM based quality management systems, you can create a best-in-class solution that offers centralized management and visibility with real-time local quality execution. With complete access to real-time data from each area of manufacturing operations, DELMIA Quality can effectively become the Quality Execution System component for third party Quality Management Systems.

DELMIA Apriso Warehouse

DELMIA Apriso Warehouse module takes the concept of collaborative supply chain execution to new heights by delivering real-time, end-to-end visibility and control that optimize industrial warehouse and logistical operations. Tying together inbound and outbound logistics with warehouse operations, DELMIA Warehouse manages cross-docking, receiving, warehousing and shipping operations, while streamlining the flow of materials throughout your extended supply chain network. DELMIA Warehouse helps reduce inventory levels, facilitate on-time deliveries, improve quality performance and accelerate inventory turnover with full material synchronization.

DELMIA Warehouse synchronizes component logistics with production activity in real-time, so materials are picked only moments before needed, or cross-docked directly from the receiving dock to production. Finished goods are racked then stored or cross-docked to the shipping dock as soon as they are completed. Full genealogy and traceability is collected from suppliers through storage, handling, production and shipping to customers. This synchronization and traceability reduces inventory and WIP, improves utilization and throughput, reduces cycle-time and cost effectively provides the visibility required to manage product quality to customers.

If you do not synchronize materials management with production execution, you cannot truly be Lean. Many of our customers have typically elected to deploy DELMIA Warehouse together with DELMIA Production to provide integrated manufacturing and logistics execution.

Warehouse Management Processes	Integrated Manufacturing and Logistics Execution Processes
• Receiving • Put Away • Picking • Hold/release • Shipping • Inventory/Cycle count • Cross Docking	• Just In Time • Just In Sequence • Kanban • Quality Inspection, Sampling • Containment • Genealogy and Traceability • Coordinates with 3rd parties

• Inter-warehouse transfers • Value-added services (sequencing, kitting, assembly, complex packaging) • RFID support	• Lean material flows

DELMIA Apriso Time & Labor

Is a browser-based, enterprise-wide portal for the real-time collection and management of time-based data. It provides time and labor processing with adaptable, user-configurable rules-based architecture that can be tailored to your specific requirements. The system also interfaces to leading ERP, financial and payroll systems. With DELMIA Apriso Time and Labor, the cycle of automation is completed by making time and labor data available for review, signoff, reporting and export to back office systems.

The Time & Labor module supports –

- Electronically tracks time and attendance for all employees, producing time cards and time sheets as well as absence and late reports
- Dynamically collects data from multiple sources, such as PCs, touch screens and barcode terminals
- Configurable calendars, shifts, work periods and breaks.
- Tracks employee qualifications / certifications
- Configurable earnings and labor codes, overtime, shift premium rules and accruals.
- Supports group scheduling
- Automates the manual time card

Reporting Analytics Platform

RAP, or the reporting analytics platform is the backend or as we like to refer to it as – "the plumbing" which is the process of getting the data out of Apriso and into a more usable format. Apriso is, and has been a system designed for execution – making sure that the people and machines in the factory know exactly what to do. We optimized the existing Apriso backend to handle this execution as fast as possible. The problem is that this structure is not designed for reporting. So we deliver with RAP a fully optimized data warehouse designed for reporting.

The process of moving the data out of the execution based Apriso and into the reporting-based Data Warehouse is called an ETL or Extract, Transform and Load. Extracting the data out of Apriso, Transforming or flattening it to the reporting format and loading this into the DW at a very fast rate. Again – this is the plumbing that must occur so that both systems are kept in synch in near real time.

An additional benefit of this is that there is no need to keep several months of data in Apriso. This can now be moved out into the RAP Data warehouse and the Apriso Execution Database can be kept much leaner resulting in faster performance and execution.

A large number of queries arise as to how we keep track of all the changed information that occurs. Data can be done from the operator screens, downloaded from the ERP, sent from a machine or even manually updated in query windows by DBAs. We used two technologies to do this. Change Tracking for SQL and Change Data Capture for Oracle. Both technologies essentially do the same thing in making sure every single update, insert and delete is recorded. This is what we send to the Data Warehouse and is an essential component to the MPI plumbing.



The diagram above explains how the two modules work with the rest of Apriso (important for customers that already have Apriso without RAP/MPI):

- First we have created the RAP data warehouse that is continuously updated with the latest data from Apriso operational database. That allows Apriso users to execute complex reports on this database without impacting Apriso performance.
- On top of that data warehouse we have implemented OLAP (Online Analytical Processing) Cubes that can be queried by Apriso services using Multidimensional Expressions (MDX) Queries (using new MDX Query function type)
- Data collected from RAP by Apriso services can be used jointly with other Apriso data and provided to end users in Apriso portal (using also the new Cube Viewer function displaying data in Pivot Table)
- RAP OLAP Cubes can be also accessed MS Excel or any 3rd party BI tools without impacting Apriso stability / performance
- MDX Query = Multidimensional Expression, used to query cubes and return multidimensional cellsets

Dashboard Builder

Easy, non-technical way of creating multi-source visually appealing dashboards displaying aggregated manufacturing data and making them universally available across the entire enterprise to facilitate access to key information through clean and tailored visualizations

The key capabilities for Dashboard Builder are –

- Self-service dashboards
  - Create shared dashboards
- Access transactional and analytics data
- For super-users, supervisors, data analysts
- Dashboards created visually (no PB involved)
- Time & skills to create dashboard – like Excel
- Global
  - Multilanguage, can be managed centrally at COE



MPI Lite

Delivered as a completely integrated DELMIA Apriso solution, MPI Lite will help reduce the complexity of performing mission critical analysis on current or historical data. Analysis can be embedded directly within DELMIA Apriso business process or performed off-line, to better identify areas for performance improvement. Results can then be easily applied to your operations processes, establishing a closed loop that supports further continuous improvement.

Dashboard & Reports will be built on the same DELMIA Apriso unified data model, so includes preconfigured data mapping, consistent KPIs and dashboards that span production, logistics, quality and maintenance operations.

The Dashboards and Reports will provide the following benefits for TRSL:

- Establish near real-time visibility and control over shop floor activities that is actionable to improve performance
- Improve coordination and efficiency across all operations
- Support continuous improvement at plant level
- Accelerate and improve decision support
- Better understand impact of events to stay on plan
- Enable non-technical users to leverage plant data
- Increase worker, line, department, and plant efficiency

DELMIA Apriso Integration Framework

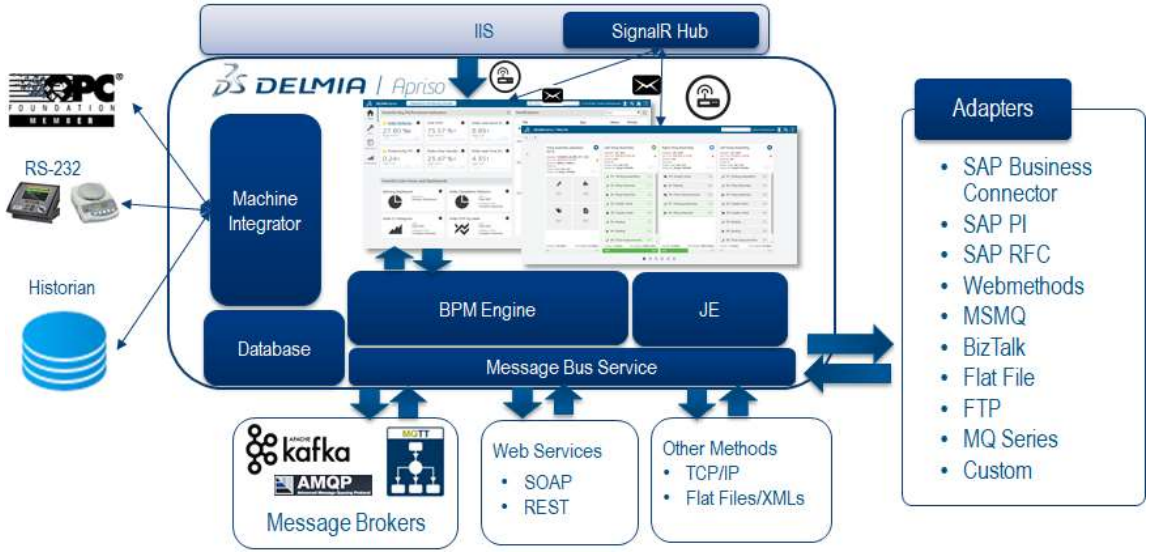
DELMIA Apriso has strong inbuilt capability to integrate with external systems. The Machine Integrator and Business Integrator modules help with integration with automation systems and IoT sensors / devices and external business systems such as ERP respectively.

TRSL will need to extend necessary support (either internally or through other partners / vendors) to enable / expose the data from existing systems and instruments so that the data can be automatically integrated with our platform.

The DELMIA Apriso Message Bus architecture consists of the following components:

- The BPM engine component dedicated to message publishing and a snippet that can subscribe or publish messages.
- The Message Bus Service installed as an additional service for the DELMIA Apriso Incoming message monitor.
- The SignalR hub, which comes as a preinstalled service used by DELMIA Apriso.
- The One DELMIA message Broker used to communicate with the 3DEXPERIENCE Platform.

- The Kafka enterprise message bus brokers.
- and the MQTT protocol brokers.



Business Integrator is the Enterprise Application Integration (EAI) component of platform, using wide-ranging, pre-built application adapters for bi-directional connectivity to multiple external systems. By leveraging web services and a Service Oriented Architecture (SOA), Business Integrator simplifies connectivity to enterprise systems. Business Integrator globally integrates planning, design and production, enabling manufacturers to become a truly adaptive enterprise. Business Integrator offers dozens of off-the-shelf interfaces for SAP, Oracle or other Enterprise Resource Planning systems, as well as 3DEXPERIENCE platform to extend the value of ERP and close the loop between PLM and Shop Floor. This integration capability enables easy adaption to existing IT infrastructure, all based on XML standards.

Business Integrator:

- Imports order details from ERP for document generation, labeling and data validation
- Controls the mapping of data to your ERP transactions
- Collects 'as built' and 'as maintained' product and process PLM information; these records may contain information about:
  - What specific parts were used in production
  - Which Bill of Process was used to create it
  - Monitoring and quality control information, as the product went through the entire manufacturing process
  - Exports and uploads transactions and process information into your ERP or 3DEXPERIENCE platform on an immediate or periodic basis
  - Automatically queues exported transactions when your ERP or 3DEXPERIENCE platform system is unavailable, avoiding the possibility of a production line stoppage based on enterprise application unavailability

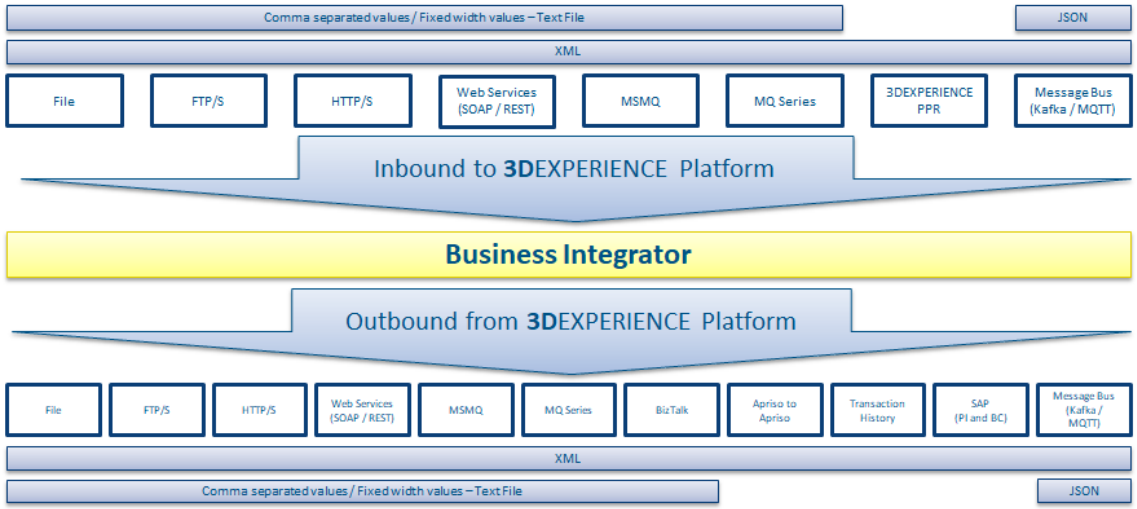
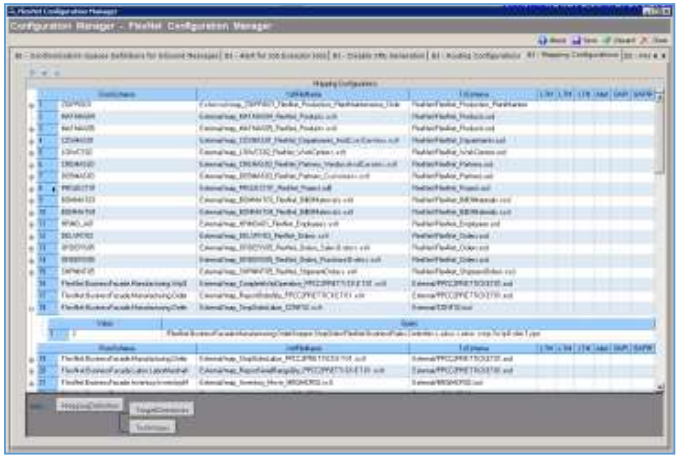


Figure 3 Business Integrator Interfaces

We have successfully implemented systems tightly integrated with SAP, Oracle, QAD, Infor, and custom ERP systems. Our platform has the widest, deepest set of ERP integration capabilities available on the market covering the full scope of manufacturing operations including production, maintenance, materials, labor and quality transactions. Business Integration will be used to interface with SAP ERP to download work orders and other master data for execution of operational processes.



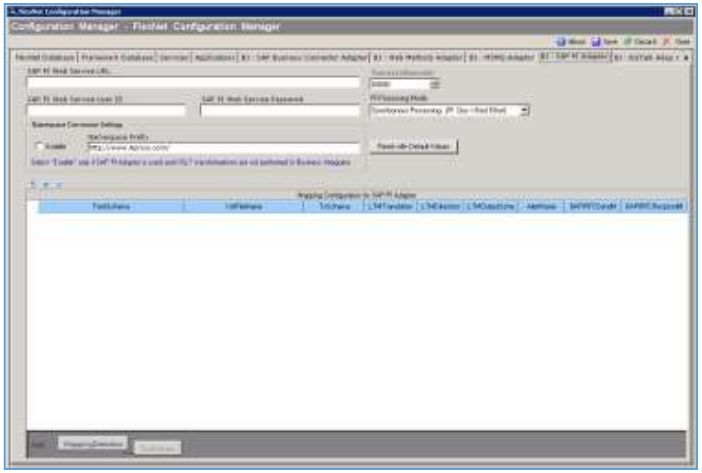
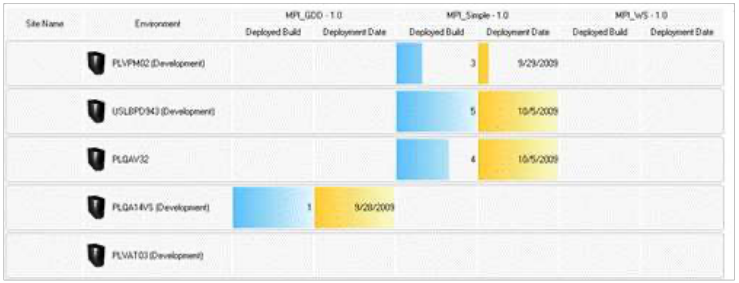
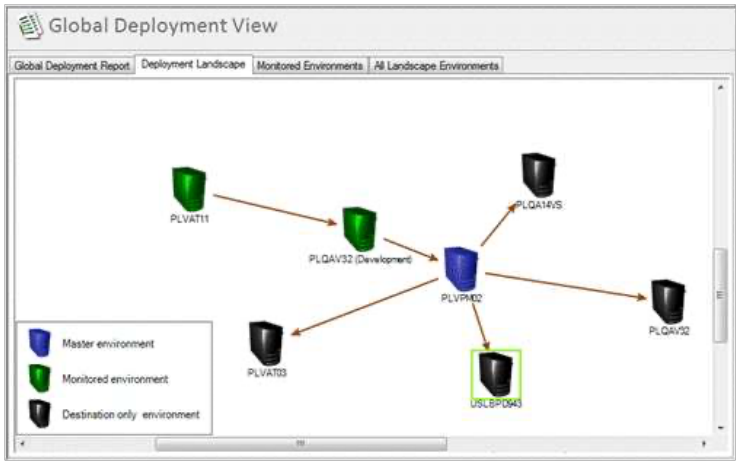


Figure 4 Business Integrator Configuration

DELMIA Apriso Process Manager

Is used to package, deploy and govern manufacturing processes across multiple plants and other facilities. GPM deploys systems configuration data and processes with logic and user interface, database objects, machines configuration and third party components. GPM can handle both complete processes and delta changes being a result of continuous improvement practices. Incremental changes can be quickly deployed with no downtown to bring immediate return without a need to retest the entire IT system. Used at COE (Center of Operational Excellence), DELMIA Global Process Manager becomes an integral component to best manage the complete lifecycle of a process. Beyond improving agility, standardizing business processes across global manufacturing and supply chain operations it helps to reduce IT costs, ensure brand integrity, ease process governance and accelerate the time to value of IT systems deployment. By implementing DELMIA Global Process Manager as the "backbone" of your COE you get real-time visibility and control over processes at every manufacturing facility – for all manufacturing operations.





2. Years of Industry Experience

Dassault Systèmes, the 3DEXPERIENCE Company, founded in 1981, and for more than 40 years, we have been providing businesses and people with virtual universes to imagine sustainable innovations. Its 3DEXPERIENCE Platform leverages the Company's world-leading 3D software applications to transform the way products are designed, produced, and supported, enabling businesses to craft delightful TRSL experiences.

In the Manufacturing Execution Systems space, Dassault Systèmes has been helping global manufacturing and industrial customers improve their operational performance for more than 20+ years. Over the years, Dassault Systèmes has matured and aligned its solution offering to the industries and respective segments with targeted industry solution experiences and processes. Please refer to the below link for more details –

https://my.3dexperience.3ds.com/welcome/compass-world/3dexperience-industries/transportation-and-mobility/lean-production-run

3. Experience in Rolling Stock Projects (please mention specific company names)



For more details refer to https://www.3ds.com/insights/customer-stories/alstom-transport-improves-industrial-operations-performance

Leading rolling stock manufacturer from China -

MANUFACTURING OPERATIONS MANAGEMENT
at leading rolling stock company in China



Customer is the No. 1 rolling stock manufacturer in the world and is the largest base of railway vehicles R&D, manufacturing, maintenance/repair and exporting in China.

DELMIA Apriso Offer

Maintenance, Production & Quality, Time & Labor

DELMIA Users

1000+ users

Customer Since

2019

Challenges

- Responding to planning change was frequently delayed
- Production confirmation was much later than real-time, and it was hard to calculate fulfillment ratio of planning
- The resources, machines, materials and quality is managed manually in the manufacturing processes

DELMIA Apriso Benefits

- Precise and quick transferring of engineering data between R&D and shop floor
- Improved fulfillment ratio of production planning
- Transparency of production progress and exceptions
- Complete vehicles production tracking and tracing
- Quick response to material shortages
- Data-based production operations management

DELMIA Apriso Capabilities

- Production Data Integration with ERP
- Production Order Management
- Enhanced Shop-floor Execution
- Quality Management, Integration with QMS
- Tracking and Tracing
- Maintenance: Machines, Tools, Jigs

33 DELMIA

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4. Major Projects / Industries Implemented, Project Background in PLM-related Industries & Background in Other Heavy Industries/Manufacturing (please mention company names), Total Number of Licenses Currently in Operation

DELMIA Apriso has a strong user base and presence across multiple industry segments and customers. Below is a very small list of customers across industries. Due to confidentiality clauses with our existing customers, we are not able to put out a full list of customers.

Today DEMIA Apriso has a very large community of users, more than 700K+ across the world, spread across NAM, EuroNorth, Japan, EuroWest, EuroCentral, China and India.



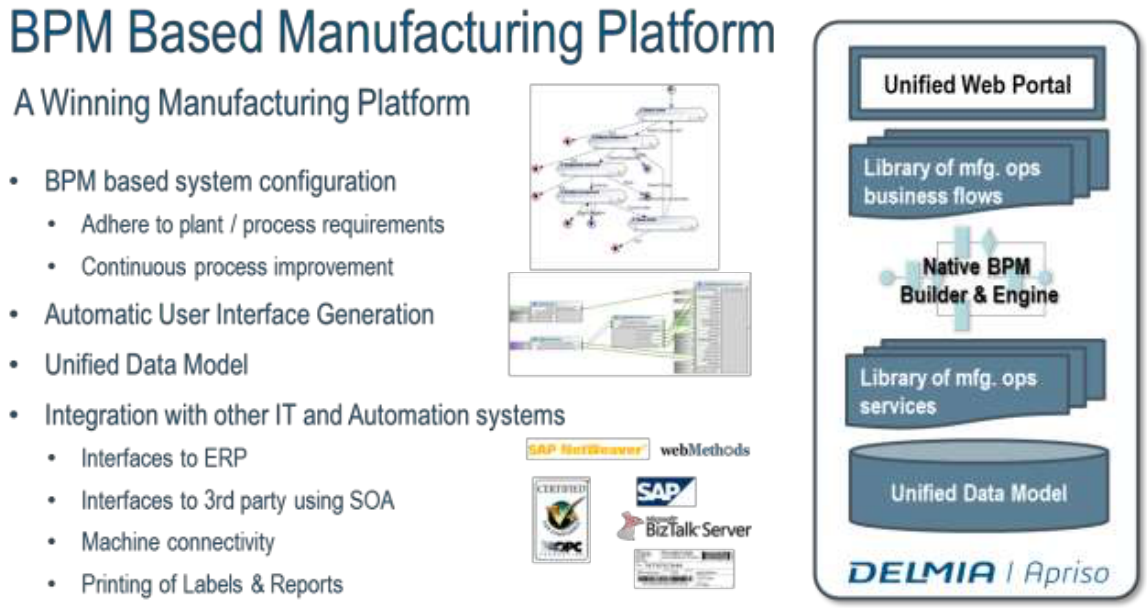
5. Integration Capability with PLM System – Windchill & Integration Capability with ERP System – SAP

Please refer to section on Business Integrator as detailed above for details around integration with PLM and ERP.

6. Customization and Flexibility of the Solution

Flexible Production with DELMIA Apriso Business Process Management

Global manufacturing leaders are those able to respond quickly to change while maintaining process consistency and quality standards. One approach they take is to establish a COE team to configure and distribute process improvements. By combining a COE with a global platform for manufacturing operations management that embeds Business Process Management (BPM), TRSL can simplify process distribution, governance and monitoring, resulting in late product and process customization and changes with less effort.



DELMIA offers a unique application that is based on cross-functional processes, which are deployed across manufacturing locations. Processes authored in DELMIA Process Builder reference all elements of manufacturing operations, including work instructions (WI), documents and characteristics, and specifications. These processes also include technical components, such as database objects, reports, label definitions, machine configuration and dictionary data. Hence, TRSL can easily and quickly push late changes in its processes (including WI, documents, etc.) to all its plants.

DELMIA Apriso BPM has the following unique capabilities:

- Single tool for authoring unique business processes.
- Supports Technical & non-technical users
- Supports integration to automation layer
- Supports integration to Business Systems (ERP, PLM, SCM, etc.)
- Supports run-time execution of business logic

- Supports automatic generation and rendering of shop floor UI without coding
- Supports a broad scope of mobile devices
- Support of Prototyping - define only the "surface" layer of a process including UI, screens, buttons, and navigation between screens
- Supports Blueprinting - allows for in-line documentation of business processes

The native BPM technology is a key differentiator for DELMIA, which makes it more adaptive to the speed of change in two ways:

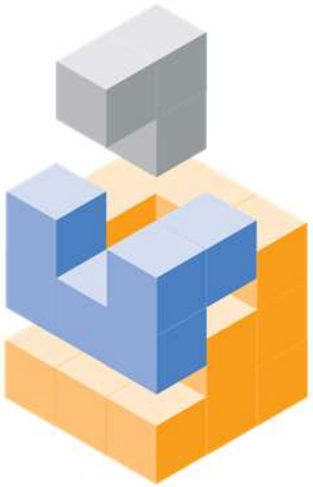
- By empowering plant-floor operations and information technology professionals to configure or reconfigure the system with a minimal amount of "configuration" and with no programming needed (no compilation, external sources, packaging).

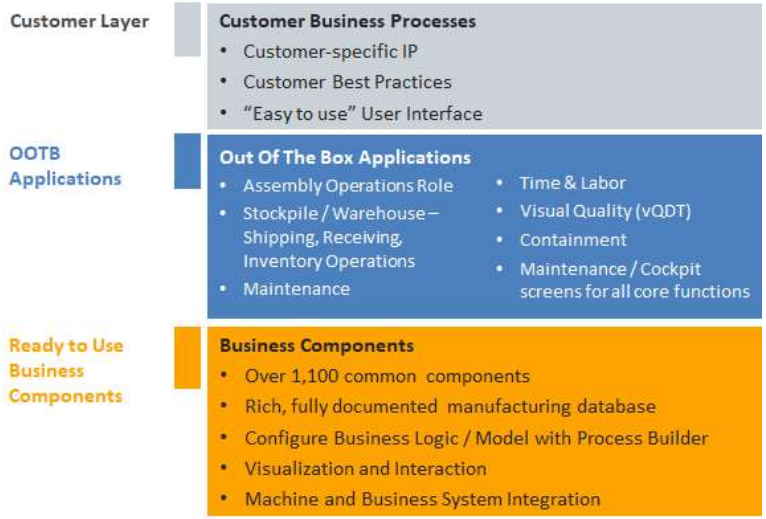
By coordinating end-to-end processes that synchronize all areas of operations, from materials, to production, quality, maintenance, and shipping.

DELMIA Apriso Platform Building Blocks

The Building Blocks are what enables us to achieve a near 100% fit to TRSL requirements.

1. You can think about it like Microsoft Excel, where orange layer is core Microsoft product you've installed on your computer. This is the bottom layer of the diagram, and represents the standard software, which provides a large number of capabilities generic to Operations Management. The standard software includes a number of engines that provide functionality around, for example, traceability, business rules, production management, materials management, visualization and reporting. This layer provides approximately 80% of the total functionality of any application.





2. The blue layer would be an "out-of-the-box" application or Role. Complex Assembly Operator, Assembly Supervisor, Warehouse Operator and Warehouse Supervisor are examples of Roles. "Cockpit" or "Maintenance and Monitoring" (M&M) screens are another example. All major databases (Products, Inventory, Maintenance, etc.) have associated M&M screens and there are functions like Containment that use the M&M format to address a particular business issue / function. This layer represents approximately 10% of the total solution. Not all the out of the box applications and capability are applicable to TRSL, the above description is general capability of the DELMIA Apriso platform.

3. What we find with our customers is that their primary competitive advantage is the Intellectual Property (IP) that defines how they do manufacturing and related functions such as quality, logistics, etc. No out-of-the-box software can accurately reflect all of this functionality since some of it is IP that is unique to each customer. Now we are in the grey level: in the Excel analogy, I have created an excel spreadsheet workbook that is unique to me. At the same time I have never touched core excel, so when I go to upgrade to the next version, it is a matter of hours or days of downtime, it is not a completely new project or customization, which you might be used to with some solutions. The top layer represents Process-Builder specific configuration. It is the part of the system that will be configured using Process Builder during a project to provide the final 20% of the solution. This part of the solution incorporates those business rules, logic and IP that is specific to your company. These configuration changes do not require any changes to the core software. This means that the core software can always be upgraded to the latest version without significantly affecting the customer-specific configuration.

The benefit of this architecture is that the customer-specific layers of the software, blue and grey, are divorced from core application. This means we can achieve near 100% fit without any customization because we are not touching core software or any of the code within it.

7. Scalability, Real-Time Data Processing, and Analytics Capabilities

Functional Scalability –

DELMIA Apriso has many other modules and capabilities that allow it to be leveraged for TRSL's continuous improvement initiatives. These include Maintenance module and Machine Integration. Even with the existing scope of modules, there are many additional use cases that can be implemented for TRSL over the course of the project and beyond. Below is the brief description of the additional modules –

DELMIA Apriso Maintenance –

DELMIA Apriso's Maintenance module works in close coordination with other Apriso applications, allowing you to more effectively manage maintenance to minimize disruptions and maximize uptime within your manufacturing environment. Gain unprecedented real-time visibility into maintenance management operations to better manage labor, spares, impact on production schedules, quality and other factors that relate to maintenance execution, which form a vital dimension when it comes to asset intensive industry.

With DELMIA Apriso Maintenance, you can maximize equipment availability by synchronizing production, quality and other activities with maintenance operations to avoid disruptions and better anticipate equipment repairs. Establish closed-loop, paperless manufacturing processes that drive preventive and reactive maintenance to keep equipment performing consistently for higher product quality and improved manufacturing efficiency.

Specific capabilities include, but are not limited to:

- Real-time, actionable dashboards for visibility into maintenance schedules, tasks, resources and assets
- Detailed asset profiles, combined with complete service BOM / parts and support descriptions
- Labor management capabilities to match skills, training and availability with work requirements
- Checklists and safety instructions
- Serial number tracking and tracing for equipment and parts
- Provide automatic reminders to prevent equipment overdue service and affect product quality.

DELMIA Apriso Machine Integrator

DELMIA Apriso Machine Integrator is an integral component of DELMIA Apriso's solution for manufacturing operations management, providing a vital link to simplify management of the equipment layer. By automating real-time data collection with a distributed, bi-directional machine or process control connectivity, DELMIA Apriso integrates your business processes with machinery using common industry protocols. Pulling machine

behavior configuration out of the machine layer gives far greater flexibility, while a distributed, n-tiered architecture provides reliability, performance, and visibility into both the functional and technical dimensions of your machine's operations.

Challenges with Traditional Automation Solution –

- Many automation and equipment vendors
- PLCs with low level programming
- Business Logic is often at the PLC layer
- No "out of the box" solution - Automation programmed to meet the process needs
- High Maintenance Costs
  - Process changes, New Product Introductions
  - require significant change
  - Replacing with new equipment is expensive
- Automation is often plant centric, Expensive to rollout to other plants

Machine Integrator is used to automate real-time data collection with a distributed, bidirectional machine or process control connectivity and integrate your business processes with machinery using common industry protocols. It facilitates data acquisition and direction for machines, tools, devices and sensors.

MI is designed to communicate seamlessly as an interface to the shop floor so that all machines can be accessed in a uniform way within the entire DELMIA product suite. Located outside the machine layer, DELMIA Machine Integrator is equipment agnostic, making it much easier to centrally administer and monitor across many or all locations taking advantage of the BPM and global deployment capabilities. Built-in mechanisms deliver distributed configuration management, better reliability and higher availability.

DELMIA Apriso Machine Integrator –

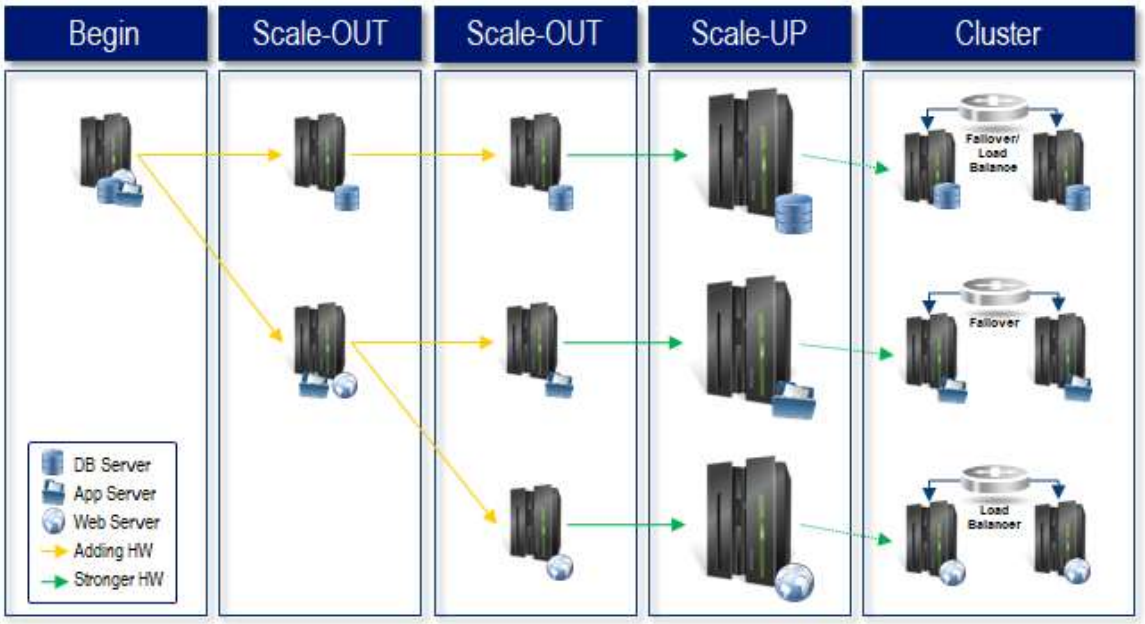
- Adheres with OPC UA and DA protocol – Most frequently used for: PLC based machinery, Robots, Palletizers & Conveyors, Test Equipment and
- RS-232 standards – Most frequently used for: Older Equipment, Scales (W&D), Torque Guns & Wrenches
- Also supports Kafka (for high volume streaming data) and MQTT (used for sensors) protocols
- Integrates tightly with Apriso's BPM engine
- Business and Process logic stored outside the PLC
- Adapts to the Business and Process Needs, Process and Product changes handled at the MES layer
- Designed for global implementations, Machine configuration and business process shared globally

Apriso is a member of the OPC Foundation and is "Compliance Certified", which is the highest level of certification possible.

The typical use cases where Apriso Machine Integrator plays a major role are listed below –

- Monitor Production
  - Collect machine events to provide performance information - KPIs, SPC
  - Receive machine status to respond to problems
- Control/Drive
  - Control machines e.g. conveyors based on the process flow
  - Send programs to machines
- Data acquisition
  - Receive measurements results

Architectural Scalability –



- **Scale-UP**

  Generally refers to adding more processors and RAM.

- **Scale-OUT**

  Generally refers to adding more servers.

- High Availability
  - Microsoft Clustering Services (WSFC) - Failover
  - Windows Network Load Balancing (NLB)
  - Readable replica
  - Real Application Cluster
  - Virtual Clustering
  - RAID for System, Application and Data Storage

- The solution is expected to evolve in time and cover more functional requirements coming from sites as they gain maturity in terms of Operations Management. Such natural evolution results in growing system load and to keep system performing it needs to be able accommodate and support this growth.
- Our solution is fully ready for long-term solutions that will grow in time. Both scaling up and scaling out scenarios are possible and can be utilized in future when such necessity will occur.

Real-time processing –

will be enabled by integration framework leveraging business integrator and machine integrator modules

Analytics Capabilities –

Will be delivered using Reporting & Analytics platform comprising of Dashboard Builder and MPI Lite modules.

8. Global Presence

Dassault Systèmes has a global presence to support customers worldwide, around three headquarters in Boston, Paris and Shanghai. We have about **25,000 passionate people** across the globe who are at the heart of the company; people sharing the same values, the same passion for innovation. We also have a large R&D team connected across 77 laboratories. Among our 25,000 people worldwide: 41% are based in Europe, 32% in Asia-Pacific and 27% in Americas. We have selected and built an ecosystem with over 14,000 partners across technology, sales and services, education, and more than 200 scientific & research partners, to increase our collective knowledge and reach.



9. Licensing Model – Subscription or Perpetual

Our licensing model is Perpetual with Annual Licensing Cost every year to support for upgrades and product bug fixes and patches

10. Local Team Presence and Support Availability



Dassault Systèmes also has a strong network of partners and system integrators who will be able to support TRSL with support and services related scope both from onsite and offshore locations.

11. Server/Cloud Infrastructure Requirements

Details already provided in the section above Proposed Hardware Sizing.
DELMIA Apriso (current version of 2025) supports the Infrastructure as a Service (IaaS) model. The IaaS infrastructure can be provided by any 3rd party vendor that supports VMWare or Hyper V virtualization.

12. Data Archiving and Storage – Timeline and Capacity

DELMIA Apriso Archiving

Is the module within the DELMIA Apriso Unified Foundation that enables to optimize the performance and structure of the database used to drive operations and measure operational performance. The DELMIA Apriso Unified Data Model captures performance and transaction data as structured information, which can then grow rapidly as new information is added or if older, unused data is not removed. DELMIA Apriso Archiving provides the robust capability of cleaning or archiving your records to maintain and optimize your data retrieval and storage system. Currently this module is not included in scope, however, it can be extended at later point of time as required by the business team. The storage of the archived data will have to be done in dedicated archival databases, which needs to be provided by TRSL. The frequency and data archival policy will be governed by TRSL business.